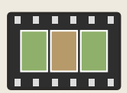
STAGE A · DATA CURATION & PERCEPTION TRAINING (built once, offline)



data

RUGD off-road runs

park · trail · creek
frame sequences



curate

Ground-ahead crops

160 px patch, ≥85%
one terrain group

label

5 terrain groups

firm · dense · rough
soft · canopy

frozen

CLIP ViT-B/32

frozen encoder
512-d embeddings

train

Bootstrap ensemble

5 linear probes
+ calibrate gate τ

frozen CLIP + probes reused at deployment

STAGE B · EMBODIED DEPLOYMENT (per camera frame, uncertainty-gated)

512-d embedding *terrain posterior* *+ disagreement*



input

Camera crop

ground-ahead
160 px patch

frozen

CLIP ViT-B/32

frozen encoder
512-d embedding

5×

Ensemble probe

5 bootstrap heads
(BALD)

Eq. 7

Epistemic gate

go / look-again /
hold (thr. τ)

Eq. 9

Utility horizon

H-step projection
margin vs switch

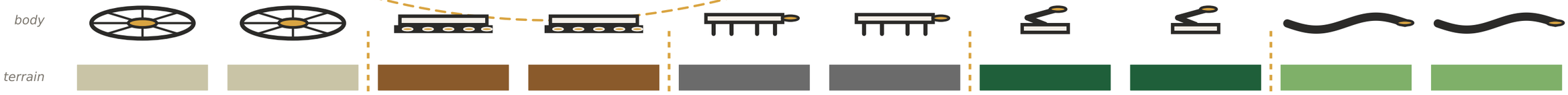
output

Morph body

reconfigure to
matched form

look-again: re-observe on high disagreement

Illustrative per-frame trace (terrain ahead → chosen body; gate fires when ambiguous):



Perception is real (camera → frozen CLIP → bootstrap ensemble); the per-body utility table is an assumed design table. Ensemble disagreement (BALD mutual information) drives both the gate and the switch margin.